

I. Mid-term Summary

Int = Intuition Bgnd = background Summary = summary
Prop = property cndn = condition

1. Convex Optimization

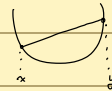
Def) Std form of cvx opt

$$\min_{x \in S} f(x) \text{ where } S = \{x \in \mathbb{R}^d : \underbrace{f_i(x) \leq 0}_{\text{inequality constraint}} \forall i \in [m], \underbrace{h_i(x) = 0}_{\text{equality constraint}} \forall i \in [p]\}$$

Above is called as cvx opt when ① $f(x)$ is cvx ftn
② S is a cvx set $\begin{cases} 1) f_i \text{ is a cvx ftn} \\ 2) h_i \text{ is an affine ftn} \end{cases}$

Def) A set S is said to be a cvx set if $\forall x, y \in S \Rightarrow \lambda x + (1-\lambda)y \in S, \forall \lambda \in [0,1]$

Def) A ftn f is said to be a cvx ftn if a) $\text{dom}(f)$ is a cvx set
b) $\forall x, y \in \text{dom}(f), \forall \lambda \in [0,1], f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$



Def) A ftn $f(x)$ is ccv if $-f(x)$ is cvx
A ftn $f(x)$ is affine if it's cvx & ccv

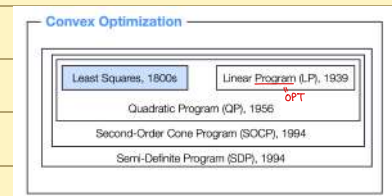
시험 변형 유형: Hw1 #3, 23기출 #3 (이전 $S, \cup S_2$ cvx? : 나옴)

Proposition) If S_1 & S_2 are cvx sets, then $S = S_1 \cap S_2$ is a cvx set.

2. Least Squares

Def) A problem is called Least Squares if

$$\min_{x \in \mathbb{R}^d} \sum_{i=1}^m (a_i^T x - b_i)^2 = \min_{x \in \mathbb{R}^d} \left\| \begin{bmatrix} a_1 \\ \vdots \\ a_m \end{bmatrix}^T x - \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix} \right\|^2 = \min_{x \in \mathbb{R}^d} \|Ax - b\|^2$$



Hopefully, it has a clld form, $x^* = (A^T A)^{-1} A^T b$

$$f(x) = \|Ax - b\|^2 = \langle Ax - b, Ax - b \rangle = x^T A^T A x - x^T A^T b - b^T A x + b^T b$$

$$= x^T A^T A x - 2b^T A x + b^T b$$

$$f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda)f(x_2) \rightarrow f(x) \text{ is } \|Ax - b\|^2 \text{ CH인 } \dots \Rightarrow \text{LHS} - \text{RHS} \leq 0$$

3. Linear Programming

Def) Std form of LP

$$\min_x \tilde{w}^T x + c \text{ where } A\tilde{x} - \tilde{b} \leq 0, \text{ i.t. } \tilde{c}\tilde{x} - \tilde{e} = 0$$

$$\begin{aligned} w, x &\in \mathbb{R}^d \\ A &\in \mathbb{R}^{m \times d} \\ b &\in \mathbb{R}^m \\ C &\in \mathbb{R}^{n \times d} \\ c &\in \mathbb{R}^p \end{aligned}$$

when ① $f(x)$ is affine
② f_i is a affine
③ h_i is an affine ftn

Props of LP: Many probs are formulated as LP ex) Resource allocation, Transportation, Linear classification prob

So, we try to make NON-LP to LP, which is called **LP Relaxation**.

ex) $p^* := \min_x w^T x : \begin{matrix} Ax - b \leq 0 \\ Cx - e = 0 \\ x_i \in \{0,1\} \quad i=1,\dots,d \end{matrix}$ Boolean Prob

$\Downarrow$

$p_{LP}^* := \min_x w^T x : \begin{matrix} Ax - b \leq 0 \\ Cx - e = 0 \\ 0 \leq x_i \leq 1, \quad i=1,\dots,d \end{matrix}$ move freedom & sometimes $p^* = p_{LP}^*$

Cons of LP: LP does **NOT** have **old form sol**
 $\Rightarrow$ ALGO (by Kantorovich) is needed

3.1. Algos for LP

1 Simplex Algo 2 Interior Point Method (IPM, based on "Strong Duality")

std form of LP: $\min w^T x : \begin{matrix} Ax - b \leq 0 \\ Cx - e = 0 \end{matrix}$

$\Downarrow$

std form of simplex algo: $\max \tilde{w}^T x : \tilde{A}x \leq \tilde{b}, x \geq 0$ $\Leftrightarrow$ slack form of LP: $\max z : A\tilde{x} = 0, x_i, s_i \geq 0, \forall i$

① $\min w^T x = \max \begin{pmatrix} w^T \\ -C^T \\ -e^T \end{pmatrix} x : \begin{matrix} Ax \leq b \\ Cx \leq e \\ Cx \geq e \end{matrix} \Rightarrow \begin{pmatrix} A \\ C \\ -C \end{pmatrix} x \leq \begin{pmatrix} b \\ e \\ -e \end{pmatrix} \Leftrightarrow \tilde{A}x \leq \tilde{b}, \tilde{x} \geq 0$

if $x_i < 0$, it can be expressed $x_i = x_{i,1} - x_{i,2}$ where $x_{i,1}, x_{i,2} \geq 0$

Q) Doesn't that expand the dimension?

A) YES! BUT we can change A : Original $Ax = a_1x_1 + a_2x_2 + \dots + a_nx_n = b$

$\rightarrow a_1(x_{i,1}) - a_1(x_{i,2}) + a_2x_2 + \dots + a_nx_n = b$ where $A' = [\tilde{a}_1, -\tilde{a}_1, \dots, \tilde{a}_n]$
 $\tilde{x}' = [x_{i,1}, x_{i,2}, \dots, x_n]$

② ex) $\max 5x_1 + 4x_2 : \begin{matrix} 3x_1 + 4x_2 \leq 78 \\ 4x_1 + x_2 \leq 36 \\ x_1, x_2 \geq 0 \end{matrix}$

$\Downarrow$

$\max z : \begin{matrix} z - 5x_1 - 4x_2 = 0 \\ 3x_1 + 4x_2 + s_1 = 0 \\ 4x_1 + x_2 + s_2 = 0 \\ x_1, x_2, s_1, s_2 \geq 0 \end{matrix}$

4. Quadratic Programming

Def) A Problem is a QP when General form of QP

$\min w^T x + \frac{1}{2}x^T Q x : \begin{matrix} Ax - b \leq 0 \\ Cx - e = 0 \end{matrix}$ where Q is **sym** ($Q = Q^T \in \mathbb{R}^{n \times n}$) and **PSD matrix** ($Q \succeq 0$)

Def) A sym. matrix Q is **PSD** if $x^T Q x \geq 0$ holds for all x

Prop) $Q \succeq 0 \Leftrightarrow$ Eigenvalues of Q are **non-negative** (pb by λ decomposition of $Q = P\Lambda P^T$)

★ Claim) obj. fn of QP $w^T x + \frac{1}{2}x^T Q x$ is **conv** in x

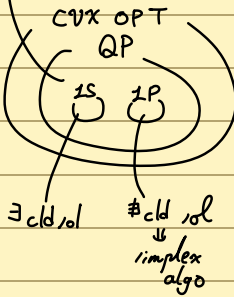
pb) $\nabla^2(w^T x + \frac{1}{2}x^T Q x) = \nabla(w^T + x^T Q) = Q \succeq 0$
 (In general, f is **conv** iff $\nabla^2 f \succeq 0$)
 $\hookrightarrow$ HW1 #6 pb)

Prop of General QP: **Strong duality** provides insights

Cons of General QP: **no old form sol**

$\exists d, d$

4.1. QP subsumes LP & LS



① QP subsumes LP : $\min_{x \in \mathbb{R}^n} w^T x + x^T Q x : Ax - b \leq 0$ reduces to LP when $Q = 0$
 $Cx - e = 0$

② QP subsumes LS : $\equiv$ special QP in which d.d.-form sol exists
 is called "Equality-constrained Least-Squares"

$$\min_{x \in \mathbb{R}^n} \|Ax - b\|^2 : Cx - e = 0 \quad \begin{matrix} A \in \mathbb{R}^{m \times d} \\ b \in \mathbb{R}^m \\ C \in \mathbb{R}^{p \times d} \\ e \in \mathbb{R}^p \end{matrix} \quad \begin{matrix} \text{(assume } m \geq d, \text{ otherwise } \neq \text{ unique sol)} \\ \text{rank}(C) = p \\ \text{rank} \begin{pmatrix} A \\ C \end{pmatrix} = d \end{matrix}$$

a) Case 1 : $p \geq d$ (방정식 # > 미지수 #, 자유도)

x^* is solely by $Cx = e$ (nth to do w/obj.)

$\Rightarrow$ Boring $\square$

b) Case 2 : $p < d$ (방정식 # < 미지수 #, 자유도)

⊕ Assume i) $\text{rank}(C) = p$: full rank $\rightarrow$ full rank p 개 제약조건이 서로 중복 x, 독립적

ii) $\text{rank} \begin{pmatrix} A \\ C \end{pmatrix} = d$: full col rank $\rightarrow$ 최적점의 유일성을 보장하는 조건

Main Thm

$$\min_{x \in \mathbb{R}^d} \|Ax - b\|^2 \quad \begin{matrix} A \in \mathbb{R}^{m \times d} \\ b \in \mathbb{R}^m \\ C \in \mathbb{R}^{p \times d} \\ e \in \mathbb{R}^p \end{matrix} \quad \text{Assumptions : } \begin{matrix} m \geq d \\ p < d \\ \text{rank}(C) = p \\ \text{rank} \begin{pmatrix} A \\ C \end{pmatrix} = d \end{matrix}$$

$\Rightarrow$ The d.d. form sol of this QP is

$$x^* = d\text{-components} \left\{ \begin{pmatrix} 2A^T A & C^T \\ C & 0 \end{pmatrix}^{-1} \begin{pmatrix} 2A^T b \\ 0 \end{pmatrix} \right\}$$

where $d\text{-components}(\cdot)$ is an operator taking the first d components

Pf Lemma 1 : If $\exists (x^*, z) \in \mathbb{R}^{d+p}$ s.t. $\begin{pmatrix} 2A^T A & C^T \\ C & 0 \end{pmatrix} \begin{pmatrix} x^* \\ z \end{pmatrix} = \begin{pmatrix} 2A^T b \\ 0 \end{pmatrix}$
 then x^* must be the opt. sol. i.e. $\|Ax - b\|^2 \geq \|Ax^* - b\|^2, \forall x$ s.t. $Cx = e$

$$\begin{aligned} \text{pf) } \|Ax - b\|^2 &= \|(Ax - Ax^*) + (Ax^* - b)\|^2 \\ &= \|Ax - Ax^*\|^2 + \|Ax^* - b\|^2 + 2(Ax - Ax^*)^T (Ax^* - b) \\ &\geq \|Ax^* - b\|^2 \quad \quad \quad = 0 \end{aligned}$$

Lemma 2 : If $\text{rank}(C) = p$ & $\text{rank} \begin{pmatrix} A \\ C \end{pmatrix} = d$,
 then $\begin{pmatrix} 2A^T A & C^T \\ C & 0 \end{pmatrix}$ is invertible.

pf) Suppose $M = \begin{pmatrix} 2A^T A & C^T \\ C & 0 \end{pmatrix}$ is NOT invertible.

then $\exists \tilde{v} = \begin{pmatrix} \tilde{x} \\ \tilde{z} \end{pmatrix} \neq 0$ s.t. $M\tilde{v} = 0$

$$\Rightarrow 2\tilde{x}^T A^T A \tilde{x} + \tilde{x}^T C^T \tilde{z} = 0 \quad \xrightarrow{=0 \because M\tilde{v}=0} \quad \begin{pmatrix} A \\ C \end{pmatrix} \tilde{x} = 0 \quad \xrightarrow{\text{rank} \begin{pmatrix} A \\ C \end{pmatrix} = d} \quad \tilde{x} = 0 \quad \xrightarrow{M\tilde{v}=0} \quad C^T \tilde{z} = 0 \quad \xrightarrow{\text{rank}(C) = p} \quad \tilde{z} = 0 \quad \xrightarrow{\tilde{v} \neq 0} \quad \tilde{v} = 0$$

$\therefore M$ is invertible $\square$

5. Second-Order Cone Program (SOCP)

Def) A SOCP is as follows:

$$\min_{x \in \mathbb{R}^d} \omega^T x : \|A_i x - b_i\| \leq c_i^T x + e_i \quad i \in \{1, 2, \dots, m\} \quad \text{where} \quad \begin{matrix} A_i \in \mathbb{R}^{k_i \times d} & F \in \mathbb{R}^{n \times d} \\ b_i \in \mathbb{R}^{k_i} & g \in \mathbb{R}^n \\ c_i \in \mathbb{R}^d & w \in \mathbb{R}^d \\ e_i \in \mathbb{R} \end{matrix}$$

(this convex) $Fx = g$ Hypersphere (in $\mathbb{R}^d$) is a convex set

$\{x : \|A_i x - b_i\| \leq c_i^T x + e_i\}$ is a convex set

Prop 2

$\|A_i x - b_i\| - c_i^T x - e_i$ is a convex ftn

Prop 1

$\|A_i x - b_i\|$ is a convex ftn

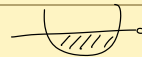
Prop 1

$\|x\|$ is a convex ftn

Prop 1: Conv preserves under addition & affine transformation

$\Leftrightarrow$ If f is conv, then $g = f(Ax+b)$ is conv

Prop 2: $\{x : f(x) \leq 0\}$ is a convex set for a convex ftn f



Def) SOC of dimension $(n+1)$ is

$$C_{n+1} = \left\{ \begin{pmatrix} x \\ t \end{pmatrix} : x \in \mathbb{R}^n, t \in \mathbb{R}, \|x\|_2 \leq t \right\}$$

$$\begin{matrix} x = A_i x_i - b_i \\ t = c_i^T x + e_i \end{matrix} \quad \text{대입해보기}$$

[Note] $\|A_i x - b_i\| \leq c_i^T x + e_i \Leftrightarrow \begin{pmatrix} A_i \\ c_i^T \end{pmatrix} x + \begin{pmatrix} -b_i \\ e_i \end{pmatrix} \in C_{k_i+1}$

i.e., "The set of x satisfying the constraint of SOCP is inverse image of the SOC C_{k_i+1} under affine mapping"

$\therefore$ SOCP is a SOC-constraint-based Program

5.1. SOCP subsumes LP & QP

SOCP: $\min_{x \in \mathbb{R}^d} \omega^T x : \|A_i x - b_i\| \leq c_i^T x + e_i \quad i \in \{1, 2, \dots, m\}$

$$F x = g$$

$$\Downarrow A_i = 0, \forall i=1, \dots, m$$

How?

LP: $\min_{x \in \mathbb{R}^d} \omega^T x : Cx \leq e, Fx = g$

QP: $\min_{x \in \mathbb{R}^d} \omega^T x + x^T Q x : Ax \leq b, Cx = e$

Rewriting QP

Recall $Q \succeq 0$. Thus we have $Q = U \Lambda U^T$

$\rightarrow$ By defining $Q^{1/2} := U \Lambda^{1/2} U^T$ & $y = Q^{1/2} x$ we have $x^T Q x = y^T y$

$$\therefore \min_x \omega^T x + x^T Q x : Ax \leq b, Cx = e$$

$$\Rightarrow \min_{x, y} \omega^T x + y^T y : Ax \leq b, Cx = e, y = Q^{1/2} x$$

still have quadratic term in cost ftn...

$\Downarrow$ Introduce another variable t

$$\min_{x, y, t} \omega^T x + t : Ax \leq b, Cx = e$$

$$y = Q^{1/2} x, \quad \underbrace{y^T y}_{\text{affine}} \leq t \quad \underbrace{y^T y}_{\text{NOT affine}}$$

$\rightarrow$ If this is SOC constraint, $\therefore$ QP is represented as SOCP format

*pb) $y^T y \leq t \Leftrightarrow 4\|y\| \leq 4t \Leftrightarrow 4\|y\|^2 + (t-1)^2 \leq (t+1)^2 \Leftrightarrow \left\| \begin{pmatrix} 2y \\ t-1 \end{pmatrix} \right\| \leq t+1$

$\therefore$ QP is a SOCP

5.2. Applications of SOCP

Practical contexts w/ uncertainty { Robust LP, Robust LS }
 When we have regularization term { Image Denoising, Penalized LS }

Robust LP

$$\min_x w^T x : \underbrace{a^T x \leq b}_{\text{random vector}} \Rightarrow \text{WTC consider } P_\sigma(\tilde{a}^T x \leq b) \geq 1 - \epsilon, \forall \epsilon > 0$$

How?

Gaussian Approximation

Assume $a \sim \mathcal{N}(\bar{a}, K)$ where $K = E[(a - \bar{a})(a - \bar{a})^T]$: Cov. Matrix
 then, we know $\tilde{a}^T x \sim \mathcal{N}(\bar{a}^T x, x^T K x)$

$$\text{Thus } P_\sigma(\tilde{a}^T x \leq b) = P_\sigma\left(\frac{\tilde{a}^T x - \bar{a}^T x}{\sqrt{x^T K x}} \leq \frac{b - \bar{a}^T x}{\sqrt{x^T K x}}\right) = \Phi\left(\frac{b - \bar{a}^T x}{\sqrt{x^T K x}}\right)$$

$\sim \mathcal{N}(0, 1)$ cdf of $\mathcal{N}(0, 1)$

$$\Rightarrow P_\sigma(\tilde{a}^T x \leq b) = \Phi\left(\frac{b - \bar{a}^T x}{\sqrt{x^T K x}}\right) \geq 1 - \epsilon \Leftrightarrow \frac{b - \bar{a}^T x}{\sqrt{x^T K x}} \geq \Phi^{-1}(1 - \epsilon)$$

$y = x^T K x$

$$\Leftrightarrow b - \bar{a}^T x \geq \Phi^{-1}(1 - \epsilon) \sqrt{x^T K x} \Leftrightarrow \|y\| \leq \frac{b - \bar{a}^T x}{\Phi^{-1}(1 - \epsilon)} \Leftrightarrow (y, \frac{b - \bar{a}^T x}{\Phi^{-1}(1 - \epsilon)}) \in C$$

i.e. $\min_x w^T x : P_\sigma(\tilde{a}^T x \leq b) \geq 1 - \epsilon$

$$\Downarrow$$

$$\min_{x, y} w^T x : y = K^{\frac{1}{2}} x, \|y\| \leq \frac{b - \bar{a}^T x}{\Phi^{-1}(1 - \epsilon)}$$

i.e. It is SOC constraint!

$\therefore$ Robust LP is SOCP

5.3. How to solve SOCP?

$$\min_{x \in \mathbb{R}^n} w^T x : \|A_i x - b_i\| \leq c_i^T x + e_i, i \in \{1, \dots, m\}$$

$$F_x = g$$

Unfortunately, $\#$ cld-form is in general. $\Rightarrow$ Rely on strong duality for (Post II) algo insights.

6. Semi-Definite Program (SDP)

Def) Linear Matrix Inequality (LMI) is an expression of the form

$$LMI(x) = A_0 + x_1 A_1 + \dots + x_m A_m \succeq 0 \quad (\text{PSD})$$

where $x = [x_1, \dots, x_m]$ is a real vector

$A_0, A_1, \dots, A_m$ are $n \times n$ symmetric matrices.

Def) A problem is JDP when

$$\min_{x \in \mathbb{R}^n} w^T x : \begin{cases} G + x_1 F_1 + \dots + x_d F_d \succeq 0 \\ Cx = e \end{cases}$$

$= LMI$: conv? hmm...

General JDP - Cons: $\#$ cld-form is in general $\Rightarrow$ Rely on strong duality

where G, F_i are symmetric

A set $S := \{x : G + x_1 F_1 + \dots + x_d F_d \succeq 0\}$ is conv $\therefore$ SDP is conv prob.

pf) Suppose $x, y \in S, \lambda \in [0, 1], \tilde{x} = [x, \dots, x_d], \tilde{y} = [y, \dots, y_d]$

$$\begin{aligned} \text{WTS } \lambda x + (1 - \lambda)y &= G + (\lambda x_1 + (1 - \lambda)y_1)F_1 + \dots + (\lambda x_d + (1 - \lambda)y_d)F_d \\ &= \lambda \{G + x_1 F_1 + \dots + x_d F_d\} + (1 - \lambda) \{G + y_1 F_1 + \dots + y_d F_d\} \succeq 0 \end{aligned}$$

$\succeq 0$ ($\because x \in S$) $\succeq 0$ ($\because y \in S$)

Claim) If A & B are PSD, then $\lambda A + (1 - \lambda)B$ is PSD for $\forall \lambda \in [0, 1]$, i.e. conv combi of PSD is also PSD

pf) $v^T (\lambda A + (1 - \lambda)B) v = \lambda v^T A v + (1 - \lambda) v^T B v \geq 0 \quad \forall v \in \mathbb{R}^n$

≥ 0 ($\because A$ is PSD) ≥ 0 ($\because B$ is PSD)

6.1. JDP subsumes LP

$$\text{SDP} \begin{cases} \min_{x \in \mathbb{R}^d} \omega^T x : G + x_1 F_1 + \dots + x_d F_d \succeq 0 \\ Cx = e \end{cases} \quad \text{where } G, F_i \text{ are symmetric}$$

$\Rightarrow$ When G, F_i are diagonal matrices, JDP reduces to an LP

($\because$ LMI reduces to $G_{ii} + x_1[F_1]_{ii} + \dots + x_d[F_d]_{ii} \geq 0, \forall i = 1, 2, \dots, m \Leftrightarrow Ax \geq b$ 형태)

[NOTE] Diagonal PSD should have non-negative elements ($\because$ PSD)

6.2. JDP subsumes SOCP

$$\text{SOCP} \begin{cases} \min_{x \in \mathbb{R}^d} \omega^T x : \|A_i x - b_i\| \leq C_i x + e_i, \quad i \in \{1, 2, \dots, m\} \\ Fx = g \end{cases}$$

Claim) $C_i^T x + e_i > 0$ w/o loss of generality

pb) o.w. $\|A_i x - b_i\| = 0$ and be merged into $Fx = g$

$$(C_i^T x + e_i)^2 \geq \|A_i x - b_i\|^2 \Leftrightarrow (C_i^T x + e_i) - (A_i x - b_i)^T \{(C_i^T x + e_i)I\}^T (A_i x - b_i) \geq 0$$

$\Updownarrow$ Schur Complement Lemma

$$F_i(x) := \begin{bmatrix} (C_i^T x + e_i)I & A_i x - b_i \\ (A_i x - b_i)^T & C_i^T x + e_i \end{bmatrix} \succeq 0$$

Schur Complement Lemma (SCL)

Suppose $A \succeq 0$, then $X = \begin{pmatrix} A & B \\ B^T & C \end{pmatrix} \succeq 0$ iff $S := C - B^T A^{-1} B \succeq 0$

cb) b is convex ftn iff $\nabla^2 f \succeq 0$
 b is strictly convex ftn iff $\nabla^2 f \succ 0$

pb) Let $b(x, y) = (x^T \ y^T) \begin{pmatrix} A & B \\ B^T & C \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = x^T A x + 2x^T B y + y^T C y$

[Note] $f(x, y)$ is strictly convex in $x \Rightarrow x^* = \arg \min_x f(x, y) = -A^{-1} B y$ ($\because A \succeq 0, \nabla^2 f = A \succ 0$)

$(\Rightarrow)$ Let $g(y) := f(x^*, y) = (A^{-1} B y)^T A (A^{-1} B y) - 2(A^{-1} B y)^T B y + y^T C y = y^T S y$

Since $X = \begin{pmatrix} A & B \\ B^T & C \end{pmatrix} \succeq 0$, $f(x, y) = (x^T \ y^T) \begin{pmatrix} A & B \\ B^T & C \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \geq 0, \forall x, y$

Thus, $g(y) = f(x^*, y) = y^T S y \geq 0, \forall y \quad \therefore S \succeq 0$

$(\Leftarrow)$ Suppose $S \succeq 0$, then $g(y) = f(x^*, y) = y^T S y \geq 0, \forall y$ where $f(x, y) = (x^T \ y^T) \begin{pmatrix} A & B \\ B^T & C \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$ and $x^* = -A^{-1} B y$

Since x^* is the minimizer of $b(x, y)$, we have $b(x, y) \geq g(y) = b(x^*, y) \geq 0, \forall x, y$

Thus, we have $X \succeq 0$

Now we know $(C_i^T x + e_i) \geq \|A_i x - b_i\| \Leftrightarrow F_i(x) := \begin{pmatrix} (C_i^T x + e_i)I & A_i x - b_i \\ (A_i x - b_i)^T & C_i^T x + e_i \end{pmatrix} \succeq 0$

WTS F_i is an LMI

$$F_i = \underbrace{\begin{pmatrix} eI & -b \\ -b^T & e \end{pmatrix}}_{\text{sym}} + \underbrace{\begin{pmatrix} (C_i^T)I & A_i x \\ x^T A_i^T & C_i^T x \end{pmatrix}}_{\text{sym}} = \begin{pmatrix} \sum C_i x_i I & \sum a_i x_i \\ \sum a_i^T x_i & \sum C_i x_i \end{pmatrix} = \sum_{i=1}^d x_i \underbrace{\begin{pmatrix} C_i I & a_i \\ a_i^T & C_i \end{pmatrix}}_{\text{sym}}$$

$\therefore F_i(x) \succeq 0$ is an LMI

Conclusion) $\min_{x \in \mathbb{R}^d} w^T x : \begin{cases} F_i(x) \geq 0 \\ \text{||} \begin{bmatrix} A_i x \\ b_i \end{bmatrix} \text{||} \leq c_i^T x + d_i \end{cases} \quad i \in \{1, 2, \dots, m\}$ is JOCP (written in the format of JDP)

$\therefore \text{JOCP} \subseteq \text{SDP}$

6.3. JDP Relaxation

ex) MAXCUT Prob

$$\max_{x_i} \sum_{i=1}^d \sum_{j=1}^d \frac{1}{2} w_{ij} (1 - x_i x_j) : x_i^2 = 1 \quad i \in \{1, \dots, d\}$$

$\Rightarrow$ Quadratic obj, eq't constraint

$\Rightarrow$ Can't solve by cvx opt. we've learned ... convert it into JDP!

$\Downarrow$ Lifting

$$p^* = \max_X \sum_{i=1}^d \sum_{j=1}^d \frac{1}{2} w_{ij} (1 - X_{ij}) : \begin{cases} X_{ii} = 1 \\ X = X^T X \succeq 0 \\ \text{rank}(X) = 1 \end{cases} \quad i \in \{1, \dots, d\}$$

affine
LM ex) $d=2, X = x_1 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + x_2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + x_3 \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$

$\Downarrow$

$$p_{\text{SDP}}^* = \max_X \sum_{i=1}^d \sum_{j=1}^d \frac{1}{2} w_{ij} (1 - X_{ij}) : \begin{cases} X_{ii} = 1 \\ X = X^T X \succeq 0 \end{cases} \quad i \in \{1, \dots, d\}$$

Obviously, $p^* \leq p_{\text{SDP}}^*$ holds.

Q, Finally, how to convert X_{SDP}^* to X^* ?

A, Note X_{SDP}^* may NOT be the form $X = x x^T$

One way to handle this issue is to apply PCA.

$\Rightarrow$ When $X_{\text{SDP}}^* = U \text{diag}(\lambda_1, \dots, \lambda_d) U^T$, we define $\tilde{X}_{\text{SDP}}^* = U \text{diag}(\lambda_1, 0, \dots, 0) U^T$ and extract X^*

7. Strong Duality

Def) A Primal Prob is $\min_x f(x) : \begin{cases} f_i(x) \leq 0 \quad \forall i \in [m] \\ Ax = b \quad \text{where } A \in \mathbb{R}^{n \times d} \end{cases}$

$$\text{A Dual Prob is } \max_{\lambda, v} g(\lambda, v) := \max_{\lambda, v} \min_{x \in \text{dom} f} L(x, \lambda, v) \quad \text{cvx in } (\lambda, v)$$

$$= \max_{\lambda, v} \min_{x \in \text{dom} f} f(x) + \sum_{i=1}^m \lambda_i f_i(x) + v^T (Ax - b)$$

Def) Lagrangian fn is $L(x, \lambda, v) := f(x) + \sum_{i=1}^m \lambda_i f_i(x) + v^T (Ax - b)$
w/ Lagrangian multipliers $\lambda \in \mathbb{R}^m, v \in \mathbb{R}^n$

Def) Let $p^* := \min_x f(x) : \begin{cases} f_i(x) \leq 0 \quad \forall i \in [m] \\ Ax = b \quad \text{where } A \in \mathbb{R}^{n \times d} \end{cases}$

$$d^* := \max_{\lambda, v} \min_{x \in \text{dom} f} f(x) + \sum_{i=1}^m \lambda_i f_i(x) + v^T (Ax - b) : \lambda \geq 0$$

then

Strong Duality means $p^* = d^*$ (Strong Duality Thm)

(Int) Solving one solves the other as well

7.1. Strong Duality & KKT

pb, ($\Rightarrow$) Suppose (1) SD holds (2) (x^*, λ^*, v^*) are optimal points
then

$$b(x^*) = p^* := \min_{x \in \text{dom } f} b(x) : b_i(x) \leq 0, \forall i \in [m]$$

$$A x = b \text{ where } A \in \mathbb{R}^{m \times n}$$

$$g(\lambda^*, v^*) = d^* := \max_{\lambda, v} \min_{x \in \text{dom } f} f(x) + \sum_{i=1}^m \lambda_i b_i(x) + v^T (A x - b) : \lambda \geq 0$$

$$= g(\lambda, v)$$

$$d^* = g(\lambda^*, v^*) = \min_{x \in \text{dom } f} f(x) + \sum_{i=1}^m \lambda_i^* b_i(x) + v^{*T} (A x^* - b)$$

$$\leq b(x^*) + \sum_{i=1}^m \lambda_i^* f_i(x^*) + v^{*T} (A x^* - b) \text{ by primary prob cdtns}$$

$$\leq b(x^*) = p^*$$

$$\text{이} = \text{이} \text{ 되려면 } \nabla_x L(x^*, \lambda^*, v^*) = 0$$

$\therefore$ Strong Duality $\Rightarrow$

$$\text{0} = \text{이} \text{ 되려면 } \lambda_i^* b_i(x^*) = 0, \forall i \in [m]$$

"KKT cdtns"

$$\begin{aligned} \nabla_x L(x^*, \lambda^*, v^*) &= 0 & \textcircled{1} \\ \lambda_i^* b_i(x^*) &= 0 & \textcircled{2} \\ b_i(x^*) &\leq 0 & \textcircled{3} \\ A x^* - b &= 0 & \textcircled{4} \\ \lambda^* &\geq 0 & \textcircled{5} \end{aligned}$$

$$\begin{aligned} (\Leftarrow) \text{ WTP \#1 } f(x^*) = g(\lambda^*, v^*) &\Rightarrow p^* \leq b(x^*) = g(\lambda^*, v^*) \leq d^* \\ \text{WTP \#2 } p^* &\geq d^* \end{aligned} \Rightarrow p^* = d^*$$

(x^*, λ^*, v^*) : any triplet satisfying KKT

$$\begin{aligned} \text{WTP \#1 : } b(x^*) &= b(x^*) + \sum \lambda_i^* b_i(x^*) + v^{*T} (A x^* - b) \quad (\text{KKT } \textcircled{2}, \textcircled{4}) \\ &= L(x^*, \lambda^*, v^*) \\ &= \min_{x \in \text{dom } f} f(x) + \sum_{i=1}^m \lambda_i^* f_i(x) + v^{*T} (A x - b) \quad (\text{KKT } \textcircled{1}) \\ &= g(\lambda^*, v^*) \end{aligned}$$

WTP #2 : Suppose (1) $\tilde{x}$ is minimizer of primal (2) $(\tilde{\lambda}, \tilde{v})$ is maximizer of dual
then

$$\begin{aligned} p^* &\stackrel{(1)}{=} b(\tilde{x}) \geq f(\tilde{x}) + \sum \tilde{\lambda}_i b_i(\tilde{x}) + \tilde{v}^T (A \tilde{x} - b) \text{ by (1)} \\ &\geq \min_{x \in \text{dom } f} f(x) + \sum \tilde{\lambda}_i b_i(x) + \tilde{v}^T (A x - b) \text{ by (1)} \\ &= g(\tilde{\lambda}, \tilde{v}) = d^* \quad \square \end{aligned}$$

7.2. KKT giving algo insights

(1) Equality-constraint LS

$$\min_{x \in \mathbb{R}^n} \|A x - b\|^2 : C x - e = 0$$

KKT cdtns

$$\begin{aligned} \nabla_x L(x^*, v^*) &= 0 \\ C x^* - e &= 0 \end{aligned}$$

Recap:

Main Thm

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \|A x - b\|^2 \\ A \in \mathbb{R}^{m \times n} \\ b \in \mathbb{R}^m \\ C \in \mathbb{R}^{p \times n} \\ e \in \mathbb{R}^p \end{aligned}$$

$$\begin{aligned} \text{Assumptions : } m &\geq d \\ p &< d \\ \text{rank}(C) &= p \\ \text{rank}([C^T]) &= d \end{aligned}$$

문제 3 참고

$$\begin{pmatrix} 2A^T A & C^T \\ C & 0 \end{pmatrix} \begin{pmatrix} x^* \\ v^* \end{pmatrix} = \begin{pmatrix} 2A^T b \\ e \end{pmatrix}$$

$\Downarrow$

This was actually KKT cdtns!

$\Rightarrow$ The dd form of the QP is

$$x^* = d\text{-components } \begin{pmatrix} 2A^T A & C^T \\ C & 0 \end{pmatrix}^{-1} \begin{pmatrix} 2A^T b \\ e \end{pmatrix}$$

where d-components (1) is an equation taking the first d components

Lemma 1 : If $\exists (x^*, v^*) \in \mathbb{R}^{n+p}$ s.t. $\begin{pmatrix} 2A^T A & C^T \\ C & 0 \end{pmatrix} \begin{pmatrix} x^* \\ v^* \end{pmatrix} = \begin{pmatrix} 2A^T b \\ e \end{pmatrix}$
then x^* must be the opt. sol. i.e. $\|A x - b\|^2 \geq \|A x^* - b\|^2, \forall x$ s.t. $C x = e$

Lemma 2 : If $\text{rank}(C) = p$ & $\text{rank}([C^T]) = d$,
then $\begin{pmatrix} 2A^T A & C^T \\ C & 0 \end{pmatrix}$ is invertible.

(2) Equality-constrained Conv Opt

primary prob $\min_x f(x) : Ax = b$ where $L(x, v) := f(x) + v^T(Ax - b)$
 $\nabla_x L(x^*, v^*) = Ax^* - b$ KKT cdt

by the 1D thm, we can just solve the dual prob

$\Rightarrow$ Dual Prob $d^* = \max_v g(v)$

$= \max_v \min_{x \in U} L(x, v)$ where $L(x, v) := f(x) + v^T(Ax - b)$ is convex in x , affine in v
unconstrained w.r.t. x

Recall!

for convex f , $x^* = \arg\min_{x \in X} f(x)$ iff $\nabla f(x^*) = 0$

$\Rightarrow$ Opt. cdt for the inner minimization prob is $\nabla_x L(x^*(v), v) = 0$ where $x^*(v) = \arg\min_{x \in X} L(x, v)$

Again, $d^* = \max_v g(v)$

$= \max_v \min_{x \in U} L(x, v)$

$\min_{x \in X} L(x, v)$ is convex in v

is unconstrained w.r.t. v

KKT cdt again!

$\Rightarrow$ Opt. cdt for outer maximization prob is $\nabla_v L(x^*(v^*), v^*) = 0$ where $L(x^*(v), v) = \min_{x \in X} L(x, v)$

Final Q) Then how do we find (x^*, v^*) ?

One way is ... Gradient Descent Ascent (GDA)

$d^* = \max_v \min_{x \in X} L(x, v)$

GDA Algo

1. Initialize $x = x^0, v = v^0$
2. For $t = 1, 2, \dots, T$
 - $x^{t+1} \leftarrow x^{(t)} - \alpha^{(t)} \nabla_x L(x^{(t)}, v^{(t)})$: Descent
 - $v^{t+1} \leftarrow v^{(t)} + \beta^{(t)} \nabla_v L(x^{(t)}, v^{(t)})$: Ascent

KKT cdt

$\nabla_x L(x^*, v^*) = 0$

$\nabla_v L(x^*, v^*) = 0$

If it turns out that GDA converges to (x^*, v^*) ,

then learning rate $\alpha^{(t)}, \beta^{(t)} > 0$ are chosen properly

II. Final Summary

— : 기말 출제 point

PO/TECH
이남훈 교수님

1. Gradient-Descent (GD)

Motiv)

Consider LSE : $\min_x b(x) = \frac{1}{2n} \|Ax - b\|^2$

→ old-school sol $x^* = (A^T A)^{-1} A^T b$

→ probs 1. Large scale : $(A^T A)^{-1} \in \mathbb{R}^{d \times d}$ costs $O(d^3)$ and it's expensive

2. Streaming data : May not be able to store full matrix A

3. Many probs do NOT have old-school sol

⇒ We need general-purpose iterative method!

Int)

$$x_{t+1} = x_t - \eta_t \nabla b(x_t)$$

- $\eta_t > 0$: step size (learning rate)

- Cost / iteration = $\nabla b(x_t)$

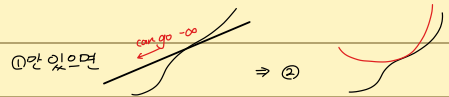
↳ ERM (Empirical Risk Mgmt) : $\nabla b(x) = \frac{1}{n} \sum \nabla b_i(x)$ and it costs $O(nd)$

limitation for big-data environment!
(회 반복마다 $n \cdot d$ 소요)

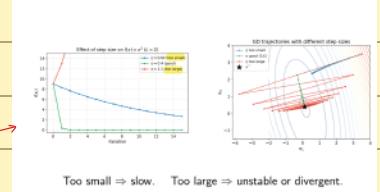
현실에서 b 는 너무 복잡
→ local approx.

$$x_{t+1} = \underset{x}{\operatorname{argmin}} \underbrace{b(x_t)}_{\text{linear approx. of } b} + \underbrace{\nabla b(x_t)^T (x - x_t)}_{\text{penalty term: don't go too far}} + \underbrace{\frac{1}{2\eta} \|x - x_t\|^2}_{\text{penalty term: don't go too far}}$$

sol Quadratics i.e., easy old-school sol.



The step size η controls everything in GD



$$\text{실제로 풀면 } \nabla b(x_t) + \frac{1}{\eta} (x - x_t) = 0 \Leftrightarrow x = x_t - \eta \nabla f(x_t)$$

⇒ Int) $-\nabla b(x_t)$: direction of greatest local decrease

η : how much can we trust local approxi. - Why? η 클수록 근사식이 넓은 범위로 맞을 것이라 예상

1.2. Pros of the Obj. : Why should we study them?

a) Convexity

b) GD finds stationary points ($\nabla f(x^*) = 0$), BUT NOT global min.

Recap) A fn is cvx if $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$, $\forall x, y, \lambda \in [0, 1]$

bin dble. $\rightarrow b(y) \geq b(x) + \nabla b(x)^T (y - x)$, $\forall x, y$

⇒ necessary cdtn for local min $x^* : \nabla f(x^*) = 0$

For cvx fns, it becomes suff. cdtn i.e., $\nabla b(x^*) = 0 \Rightarrow x^*$ is global min.

cf, what about DL?

Neural-network losses are non-cvx

Empirically GD on overparametrical models often behaves as if the probs were nearly cvx.

즉, sol을 죽인 안함아도 좋은 결과의 기준 지시

13. 1-smoothness: How to choose the step size?

Def, A fcn $f: \mathbb{R}^d \rightarrow \mathbb{R}$ is 1-smooth if its gradient is Lipschitz conti

$$\|\nabla f(x) - \nabla f(y)\| \leq L \|x - y\| \quad \forall x, y$$

Intt, gradient doesn't change rapidly

Equivl,

$$f(y) \leq f(x) + \nabla f(x)^T (y - x) + \frac{L}{2} \|y - x\|^2 \quad \forall x, y$$

control how tight the bound is

i.e., f is bdd above by a quadratic at every point

Relation w/ GD) If f is 1-smooth, then at x_t

$$f(y) \leq \underbrace{f(x_t) + \nabla f(x_t)^T (y - x_t) + \frac{L}{2} \|y - x_t\|^2}_{\text{Min. over } y \Rightarrow y^* = x_t - \frac{1}{L} \nabla f(x_t) \Leftrightarrow \text{GD w/ } \eta = 1/L}$$

$\Rightarrow$ GD w/ $\eta = 1/L$ minimizes the tightest quadratic upper bound of f .

Descent Lemma

For 1-smooth f , GD w/ $\eta = 1/L$ guarantees

$$f(x_{t+1}) \leq f(x_t) - \frac{1}{2L} \|\nabla f(x_t)\|^2$$

Intt) Guaranteed Descent: $\nabla f(x_t) \neq 0$ 이면 무조건 하강

Larger Gradient, Larger Decrease

Gradient self-tunes: Near min, $\|\nabla f(x_t)\| \rightarrow 0$ meaning we don't have to manually reduce step size.

Application) a) Suboptimality is held both ways

For 1-smooth conv f w/ minimizes x^*

$$\frac{1}{2L} \|\nabla f(x)\|^2 \leq f(x) - f(x^*) \leq \frac{L}{2} \|x - x^*\|^2$$

Gradient norm (from Descent Lemma) distance to the optimum

b) If f is conv & 1-smooth

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq \frac{1}{L} \|\nabla f(x) - \nabla f(y)\|^2$$

Intt, ∇f of 2 points are NOT just close (Lipschitz) BUT the diff. is aligned w/ $x - y$.

1-mg)

For conv 1-smooth f :

- i) Lipschitz gradient: $\|\nabla f(x) - \nabla f(y)\| \leq L \|x - y\|$
- ii) Quadratic upper bound: $f(y) \leq f(x) + \nabla f(x)^T (y - x) + \frac{L}{2} \|y - x\|^2$
- iii) Suboptimality bound: $\frac{1}{2L} \|\nabla f(x)\|^2 \leq f(x) - f(x^*) \leq \frac{L}{2} \|x - x^*\|^2$
- iv) Co-convexity: $\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq \frac{1}{L} \|\nabla f(x) - \nabla f(y)\|^2$

Thm)

Let f be conv & 1-smooth, optimality constant $\rightarrow 0$ as $T \rightarrow \infty$

then GD w/ $\eta = 1/L$ satisfies: $f(x_T) - f(x^*) \leq \frac{1}{2L} \frac{\|x_0 - x^*\|^2}{T}$ sublinear Convergence $\rightarrow$ Need $T = O(1/\epsilon)$ iterations for ϵ -accuracy

Intt) Descent Lemma가 단리적 보장이었다면 이는 $t=T$ 및 수렴 속도는 수학적으로 증명한 정리

cb) for non-conv fctns, $\min_{0 \leq t \leq T} \|\nabla f(x_t)\|^2 \leq \frac{2L(f(x_0) - f^*)}{T}$ i.e., GD finds an approximate stationary point ($\|\nabla f\|^2 \rightarrow 0, \text{ NOT } f \rightarrow f^*$)

1.4. Strong Convexity : Can GD converge faster?

Def) A diff. fcn f is μ -strongly cvx ($\mu > 0$) if

$$f(y) \geq f(x) + \nabla f(x)^T(y-x) + \frac{\mu}{2} \|y-x\|^2 \quad \forall x, y$$

Intt) f curves at least as much as a quadratic $\frac{\mu}{2} \|x\|^2$

Prop) Unique global min x^* exists

$f(x) - f(x^*) \geq \frac{\mu}{2} \|x - x^*\|^2$: grow quadratically away from x^* $\longleftrightarrow$ cvx fcn (ex. 1×1) can be flat, NO Min curvature

$\|\nabla f(x)\|^2 \geq 2\mu (f(x) - f(x^*))$: Gradient is larger whenever we are far from the optimum.

$\hookrightarrow$ Descent Lemma 대입하고 정리하면 $f(x_{t+1}) - f(x^*) \leq (1 - \frac{\mu}{2L})(f(x_t) - f(x^*))$ 즉, $\text{오차}_t \leq (1 - \frac{\mu}{2L})^t \text{오차}_0$: Exp. Convergence

$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq \mu \|x - y\|^2$: Gradient diff. is aligned w/ $x - y$, 필요 다름 필요 X

(Linear " 오차 감소)

Imp) For μ -strongly cvx f ,
 (1-smooth prop. w/ inequality reversed)
 a) Quadratic lower bound : $f(y) \geq f(x) + \nabla f(x)^T(y-x) + \frac{\mu}{2} \|y-x\|^2$
 b) Suboptimality bound : $\frac{\mu}{2} \|x - x^*\|^2 \leq f(x) - f(x^*) \leq \frac{1}{2\mu} \|\nabla f(x)\|^2$
 c) Coconvexity : $\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq \mu \|x - y\|^2$

Thm : Linear Convergence) Let f be 1-smooth & μ -strongly cvx,

then GD w/ $\eta = 1/L$ satisfies : $f(x_t) - f(x^*) \leq (1 - \frac{\mu}{2L})^t (f(x_0) - f(x^*))$

Rate : linear convergence (exp. decrease in error)

- Much faster than $O(1/t)$ for cvx fcn

- To achieve ϵ error : $T = O(\frac{1}{\mu} \log \frac{1}{\epsilon})$ iterations

1.5. Cdt# : How hard is the prob for GD?

Bgnd) For smoothness + strong convexity,

$$\frac{\mu}{2} \|x - x^*\|^2 \leq f(x) - f(x^*) \leq \frac{L}{2} \|x - x^*\|^2$$

함수 곡고 계산키로 나옴.

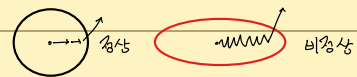
Def) Ratio btw 2 bounds : $\kappa = L/\mu (\geq 1)$ is cdt# of f .

$\kappa = 1$: perfect cdt

$\kappa > 1$: ill-cdtneel $\Rightarrow$ Intt) 강볼록성이 성립해도 목적함수가 줄은 환경에 없다면 제대로 작동 X

Application) For Quadratics $f(x) = \frac{1}{2} x^T A x - b^T x$ w/ $A \succ 0$

: $L = \lambda_{\max}(A)$, $\mu = \lambda_{\min}(A)$, $\kappa = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)}$ measure how stretched the contours are & convergence speed



2. Accelerated Methods

Bgnd) Is GD the fastest possible ALGO that only use gradients?

Consider first-order oracle model that returns $(f(x_t), \nabla f(x_t))$.

Additionally, $x_t \in \text{span} \{\nabla f(x_0), \nabla f(x_1), \dots, \nabla f(x_{t-1})\}$

$\Rightarrow$ Prob) We don't know whether this ALGO is the abs. speed limit of OPT.

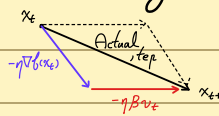
Thm: Oracle Lower Bound) For any first-order ALGO and any step $t \leq \frac{d-1}{2}$ (d: dimension),

$$\exists 1\text{-smooth convex ftn } f \text{ s.t. } f(x_t) - f(x^*) \geq \frac{32 \|x - x_0\|^2}{32(t+1)^2} : \text{dr. theoretical limit of FGDpt}$$

$\Rightarrow$ For smooth convex ftn the best possible convergence rate is $\Omega(1/t^2)$ $\neq O(1/t)$: GD
 수학적 하한선이란 의미 $\rightarrow$ 현실적 문제, GD is suboptimal!

2.1. Polyak's Momentum

Concept)



Auxiliary velocity vector (momentum) v_t which accumulates historical gradients.

Def)

$$\begin{cases} v_{t+1} = \beta v_t + \nabla f(x_t) \\ x_{t+1} = x_t - \eta v_{t+1} \end{cases}$$

η : learning rate, β : momentum coefficient, < 1

(Int) ∇ 가 같은 방향 가리키면 v_t 증가, 아니면 (= oscillate 변동) v_t 가 여제

$$\rightarrow v_1 = \nabla f(x_0), v_2 = \nabla f(x_0) + \beta \nabla f(x_0) \dots$$

$\Rightarrow v_{t+1} = \sum_{i=0}^t \beta^{t-i} \nabla f(x_i)$: Current update is weighted sum of past gradients & old ones are down-weighted

Proof)

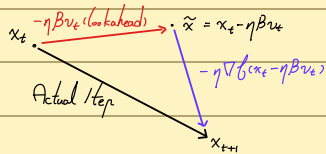
a) This method is blind. It can easily overshoot min. & roll up the other side.

b) It calculates the gradient at current position & blindly applies the stored momentum.

$\Rightarrow$ 앞을 안 보고 달리는 경주마인 심

2.2. Nesterov's Accelerated Gradient Descent

Concept)



Def)

$$\begin{cases} v_{t+1} = \beta v_t + \nabla f(x_t - \eta \beta v_t) \\ x_{t+1} = x_t - \eta v_{t+1} \end{cases}$$

(Int) Looking ahead allows the optimizer to slow down early if it's about to roll up a hill

Nesterov Convergence
 thm: 1-smooth convex

w/ carefully chosen, time-varying momentum parameter $\beta_t = \frac{t-1}{t+2}$,

Nesterov's Accelerated Gradient Descent (NAGD) achieves: $f(x_t) - f(x^*) \leq O(1/t^2)$

NAGD is theoretically optimal

" : 1-smooth strongly convex

For 1-smooth & μ -strongly convex ftn w/ cdtn # $\kappa = 1/\mu$,

NAGD achieves: $f(x_t) - f(x^*) \leq O((1 - \frac{1}{\kappa})^t)$

$\rightarrow$ JK dependence allows NAG to converge ill-ctdnd $\leftrightarrow$ SGD has linear rate of $O((1 - \frac{1}{\kappa})^t)$
 (landscapes)

$\rightarrow$ Momentum coeff. β should be adequate (large: overshoot $\leftrightarrow$ small: ineffective damping)

cb) 용어 정리:

3. Adaptive Methods

Bgnd)

Mathematical view of ill-ctdnd

$$\rightarrow \kappa = \frac{\lambda_{\max} \text{ of } \nabla^2 f(x)}{\lambda_{\min} \text{ "}}$$

$\rightarrow$ For stability (no div), $\eta \leq 2/\lambda_{\max}$ BUT progress in the flat direction is $\lambda_{\min}$

$\Rightarrow$ max. progress = $\frac{2}{\lambda_{\max}} \cdot \lambda_{\min} = \frac{2}{\kappa}$: η 하나 모든 걸 통제하려 하니 전진이 마비되며 그 정도는 κ 에 비례한다

1. 두 개념의 서로 다른 분류 속
 최적화 알고리즘을 발전시킬 때, 연구자들은 크게 두 가지 다른 접근을 합니다.
 즉 Q: 그레디언트를 구할 때 "데리티브를 얼마나 쓸 것인가?"
 이 혹은 한 수단을 찾기 위해 "가중치를 계산하는 대칭(대칭)의 양"을 기준으로 나눕니다. (올바른 8
 페이지 내용)
 • GD (Gradient Descent): 한 걸음 가기 위해 "현재 데리티브(가중치)의 기울기를 다 계산합니다.
 결핵이지만 대단히 무겁습니다.
 • SGD (Stochastic Gradient Descent): 한 걸음 가기 위해 "무작위로 샘플링한 데이터 1개(또
 는 일부 미니배치)"의 기울기를 계산합니다. 배드지만 노이즈가 심해 예측불확하게 움직입니다.
 즉 Q: 계산된 그레디언트로 "스텝을 어떻게 업데이트할 것인가?"
 이 혹은 데리브 정도 상관없이, 기울기가 주어졌을 때 물리적으로 움직이는 패턴(운동)을 기준으로 나눕니다.
 (올바른 15, 16페이지 내용)
 • Standard (표준화): 관심 없이, 그냥 현재 위치의 기울기 방향으로 곧장 이동합니다.
 • Polyak's Momentum (표준화): 초기에 이동했던 방향과 속도를 가미해 두었다가, 현재 기울기에
 "불러먹힌 관성(inertia)"을 더해 가속도를 높입니다.
 2. 조합에 따른 올바른 명칭
 따라서 이 두 가지 혹은 서로 독립적이기 때문에 다음과 같이 조합하여 서용합니다.
 • GD + 표준화 업데이트 = Standard GD (우리가 맨 처음 배운 경사하강법)
 • GD + Polyak 모멘텀 = GD with Momentum (또는 Heavy Ball Method)
 • SGD + 표준화 업데이트 = Standard SGD (매 스텝 데이터 일부만 쓰지, 관성도 없는 방법)
 • SGD + Polyak 모멘텀 = SGD with Momentum (SGDM)
 즉, 일반적으로 "Polyak's momentum을 적용한 것"은 GD에도 적용할 수 있고 SGD에도 적용할 수 있는
 가속 기법일 뿐입니다.

3.1. 1d₍₁₎ : AdaGrad

Idea) "Keep a running sum of the squares of a historical gradients for each parameter"

Def) Let $g_t = \nabla \ell(x_t)$.
 Accumulate square gradients : $G_t = G_{t-1} + g_t \odot g_t$
 & Update rule : $x_{t+1} = x_t - \frac{\eta}{\sqrt{G_t + \epsilon}} \odot g_t$

Limitations) G_t only accumulate positive values over time ($\because g_t \odot g_t$)
 $\rightarrow$ learning rate $\frac{\eta}{\sqrt{G_t + \epsilon}} \rightarrow 0$ before reaching opt sol.

직접 작성
문제풀이

3.2. 1d₍₂₎ : RMSProp

Def) Accumulate square gradients w/ decay parameter β
 $v_t = \beta v_{t-1} + (1-\beta)(g_t \odot g_t)$: EMA of gradients
 & Update rule : $x_{t+1} = x_t - \frac{\eta}{\sqrt{v_t + \epsilon}} \odot g_t$

3.3. 1d₍₃₎ : AdaM ~~XXXX~~ : Adaptive Moment Estimation = Momentum + RMSProp

Def) First moment : $m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t$: EMA of gradients
 Second moment : $v_t = \beta_2 v_{t-1} + (1-\beta_2)(g_t \odot g_t)$: EMA of squared gradients
 Update Rule : $x_{t+1} = x_t - \frac{\eta}{\sqrt{v_t + \epsilon}} \odot m_t$

Proof) $m_0 = 0, v_0 = 0 \rightarrow m_t$ & v_t are heavily biased towards 0 ($\because \beta_1, \beta_2 \approx 1$)

1d) Bias correction formula : $\hat{m}_t = \frac{m_t}{1-\beta_1^t}, \hat{v}_t = \frac{v_t}{1-\beta_2^t}$

$\Rightarrow$ Final Adam Update

step 1. Compute gradients : g_t

step 2. Update biased moments : m_t, v_t

step 3. Compute bias-corrected moments : $\hat{m}_t, \hat{v}_t$

step 4. Apply Updates : $x_{t+1} = x_t - \frac{\eta}{\sqrt{\hat{v}_t + \epsilon}} \odot \hat{m}_t$

Proof) No matter how massive the raw gradient g_t is, update magnitude is strictly held by η .
 $\frac{\eta \cdot m_t}{\sqrt{v_t}} \approx \frac{\eta \cdot g_t}{\sqrt{g_t^2}} = \eta \cdot \text{sign}(g_t)$

cb) Adam does NOT converge for all convex problems.

$\rightarrow$ 1d) AMSGrad : $\hat{v}_t = \max(\hat{v}_{t-1}, v_t)$

$\hookrightarrow$ theoretically superior BUT Adam often empirically outperforms AMSGrad.

일종의 heuristic

cb) L2 Regularization : $\ell_{\text{reg}}(x) = \ell(x) + \frac{\lambda}{2} \|x\|^2 \Rightarrow \nabla \ell_{\text{reg}}(x) = \nabla \ell(x) + \lambda x$

Weight Decay : $x_{t+1} = (1-\eta\lambda)x_t - \eta \nabla \ell(x_t)$

$\rightarrow$ In 1d GD, these 2 regularization techniques are mathematically identical.

BUT If used w/ Adam, penalty λx_t is added before scaling, leading to probs.

⇒ sol) Adam w/ 12 Reg : $x_{t+1} \approx x_t - \eta_t \odot \hat{m}_{t, \text{grad}} - \eta_t \odot \lambda x_t$
 Adam w/ Weight Decay : $x_{t+1} = x_t - \eta_t \odot \hat{m}_{t, \text{grad}} - \eta_t \lambda x_t$

3.4. Limitations of Diagonal Preconditioning

- Adam & RMSProp use diagonal preconditioning (element-wise scaling)
 - Assumption: All parameters in the network are ind.
 - Reality: Neural network parameters are highly correlated
 - Id: Capture these off-diagonal correlations using Hessian matrix
- ⇒ SQM Methods!

//

4. Second-Order Methods

4.1. Newton's Method: Minimizing second-order approximation

Recap) GD = linear approxi. + "don't go too far"

$$x_{t+1} = \underset{x}{\text{argmin}} \ell(x_t) + \nabla \ell(x_t)^T (x - x_t) + \frac{1}{2\eta} \|x - x_t\|^2$$

$$= \frac{1}{2} (x - x_t) \left(\frac{1}{\eta} I \right) (x - x_t)$$

⇒ GD assumes the same curvature in every direction (isotropic)

Q) What if we use the actual quadratic approxi?

GD : $\ell(x+d) \approx \ell(x) + \nabla \ell(x)^T d + \frac{1}{2} d^T \frac{1}{\eta} I d$
 Newton : $\ell(x+d) \approx \ell(x) + \nabla \ell(x)^T d + \frac{1}{2} d^T \overset{\text{real curvature}}{\nabla^2 \ell(x)} d$

1-order 2-order
Taylor expansion Taylor expansion

$$\overset{(1)'}{\nabla \ell(x) + \nabla^2 \ell(x) d = 0} \Leftrightarrow d = -(\nabla^2 \ell(x))^{-1} \nabla \ell(x), \quad x_{t+1} = x_t - (\nabla^2 \ell(x))^{-1} \nabla \ell(x)$$

4.2. Alternative Interpretation: Newton = GD in a perfectly-conditioned space

Recap) GD conv rate = $O(K \log^{1/\epsilon})$ where $K = 1/\mu$ & $K > 1$: zig-zag
 ⇒ Can we define affine transformation s.t. the bth is locally well-conditioned?
 즉, 좋은 자질을 변형시켜 well-condition 하게 만드는 것, $f(x) \rightarrow \ell(Ay)$

⇒ Pooh let: Define $x = Ay, \ell(Ay) = g(y)$

WTFind p.d. affine transformation s.t. $\nabla^2 g(y) = I$

$$\rightarrow \nabla^2 g(y) = A^T \nabla^2 \ell(Ay) A = I$$

$$\rightarrow \nabla^2 f(Ay) = (A^T)^T A^T$$

$$\rightarrow A = (\nabla^2 \ell(Ay))^{-1/2}$$

⇒ GD 대입 : $y_{t+1} = y_t - \eta \nabla g(y_t)$

$$\rightarrow A^T x_{t+1} = A^T x_t - \eta A^T \nabla \ell(x_t)$$

$$\rightarrow x_{t+1} = x_t - \eta A A^T \nabla \ell(x_t)$$

$$\rightarrow x_{t+1} = x_t - \eta ((\nabla^2 \ell(Ay))^{-1/2})^T (\nabla^2 \ell(Ay))^{-1/2} \nabla \ell(x_t)$$

$$\rightarrow x_{t+1} = x_t - \eta (\nabla^2 \ell(Ay))^{-1} \nabla \ell(x_t) : \text{Newton's Method}$$

⇒ Newton is GD under transformed $K=1$ pooh

작업 증명해보기
 시험 출제
 (+ Newton Method 작성)

4.3 Convergence Analysis: When does Newton converge quadratically?

Bgnd)

Pure Newton can overshoot too from x^*

Why? B/c it's only local

$$f(x_t + d) = \underbrace{f(x_t) + \nabla f(x_t)^T d + \frac{1}{2} d^T \nabla^2 f(x_t) d}_{= f(x_t + d)} + \underbrace{O(\|d\|^3)}_{\text{머디지 나옴? 그동안 무시했던 3차 이상 Taylor Expansion. d 카탈을 무시하기 힘들어짐}}$$

→ can grow error & overshoot

⇒ d_t keep direction d_t , let step size $\eta_t \in [0, 1]$ ($\eta_t = 1$: pure Newton)
 < 1 : damped "

$$x_{t+1} = x_t + \eta_t d_t \quad \text{where } d_t = -(\nabla^2 f(x_t))^{-1} \nabla f(x_t)$$

Armijo cdtn)

$$f(x_t + \eta d_t) \leq f(x_t) + \alpha \eta \nabla f(x_t)^T d_t, \quad 0 < \alpha < 1/2$$

: Require f to drop by at least a fraction α of predicted slope

Intf, 어느 정도로 함수값이 줄어야 좋은거지? 에 대한 답: Armijo Cdtn

ALGO: BTLS
Backtracking Line Search

1. Choose $\alpha \in (0, 1/2)$, $\beta \in (0, 1)$, $\eta = 1$

$$2. d \leftarrow -(\nabla^2 f(x_t))^{-1} \nabla f(x_t)$$

$$3. \text{while } f(x_t + \eta d) > f(x_t) + \alpha \eta \nabla f(x_t)^T d \quad \text{do}$$

$$4. \eta \leftarrow \beta \eta \quad \leftarrow \text{Armijo 조건 밖이면 } \eta \text{ 줄이기}$$

5. end while

Newton w/BTLS)

Phase 1: Damped

Phase 2: Undamped

조건

x^* 에서 멀

x^* 에 가까

$$\text{수식 } x_{t+1} = x_t - \eta_t (\nabla^2 f(x_t))^{-1} \nabla f(x_t)$$

$$x_{t+1} = x_t - (\nabla^2 f(x_t))^{-1} \nabla f(x_t)$$

BTLS 선택

오버슈팅 방지 위해 η_t 선택

오히려 $\Rightarrow \eta_t = 1$ return

수렴

Armijo 조건 따라 꾸준히 하강

Quadratic Convergence

Convergence thm)

Let f be m -strongly cvx, M -smooth w/ L -Lipschitz Hessian.

$$\exists \lambda \leq m/2, \gamma > 0 \text{ s.t. } \left(\begin{array}{l} \text{Phase 1 } (\|\nabla f(x_t)\| \geq \lambda) : f(x_{t+1}) \leq f(x_t) - \gamma \\ \text{Phase 2 } (\|\nabla f(x_t)\| < \lambda) : \|x_{t+1} - x^*\| \leq \frac{1}{m} \|x_t - x^*\|^2 \end{array} \right.$$

$$\hookrightarrow \text{pb) a) Lipschitz Hessian: } \|\nabla^2 f(x_{t+1})\| \leq \frac{L}{2} \|x_{t+1} - x_t\|$$

$$\text{b) Strong Convexity: } \|x_{t+1} - x_t\|^2 \leq \frac{1}{m} \|\nabla f(x_{t+1})\|^2$$

$$\text{c) Combine: } \|x_{t+1} - x^*\|^2 \leq \frac{1}{2m} \|x_{t+1} - x^*\|^2$$

GD

Newton

/mry)

계산량

Vector addition

Vector addition + Matrix inversion & multiplication

$$O(n) \text{ or } O(n \cdot \frac{p}{\alpha} \log(\frac{1}{\epsilon}))$$

$$O(n^3) \text{ or } O(n^3 \cdot \log \log(\frac{1}{\epsilon}))$$

적외

Large push + Adequate Accuracy

Small Push + High Accuracy

4.4 Quasi-Newton Methods: Approximate Hessian Cheaply

Bgnd)

$$x_{t+1} = x_t - \eta H_t \nabla f(x_t) \quad H_t = I : \text{GD}$$

$$H_t = (\nabla^2 f(x_t))^{-1} : \text{Newton}$$

$$H_t = \text{diag}(\nabla^2 f(x_t))^{-1} : \text{Cheap (O(d)) BUT misses off-diagonal coupling}$$

비대각성분의 무시

BFGS)
Broyden-Fletcher-Goldfarb-Shanno

$$\Rightarrow x_{k+1} = x_k - \eta_k B_k^{-1} \nabla f(x_k) : \text{Gradients along the trajectory tells us what the loss looks like}$$

Intf, Hessian 직접 계산 않고 ∇f 의 변화량은 누적시켜 $B_k \approx H_k$ 를 점진하게 조립!

$$\text{surrogate model: } m_k(d) = f(x_k) + \nabla f(x_k)^T d + \frac{1}{2} d^T B_k d \approx f(x_k + d)$$

가상

→ We want $m_{k+1}d$ s.t. $\begin{cases} \nabla m_{k+1}|_{d=0} = \nabla \ell(x_{k+1}) : \nabla \text{ at new point} \\ \nabla m_{k+1}|_{d=-\eta_k d_k} = \nabla \ell(x_k) : \nabla \text{ at prior point} \end{cases}$

→ $m_k(d) = \ell(x_k) + \nabla \ell(x_k)^T d + \frac{1}{2} d^T B_k d$

s.t. $\begin{cases} \nabla m_{k+1}(0) = \nabla \ell(x_{k+1}) \quad (0) \\ \nabla m_{k+1}(-\eta_k d_k) = \nabla \ell(x_{k+1}) - \eta_k B_{k+1} d_k = \nabla \ell(x_k) \end{cases}$

$B_k(x_{k+1} - x_k) = \nabla \ell(x_{k+1}) - \nabla \ell(x_k) \Leftrightarrow B_{k+1} s_k = y_k$ (recant ξ_k)
유치/변화량 벡터 기준/변화량 벡터

→ $B_{k+1} = \arg\min_B \|B - B_k\|$ s.t. $B = B^T, B s_k = y_k$

$\Rightarrow H_{k+1} = \arg\min_H \|H - H_k\|$ s.t. $H = H^T, s_k = H y_k$

BFGS: $H_k \approx B_k^{-1}$ thus $x_{k+1} = x_k - \eta_k H_k \nabla \ell(x_k)$

cf, BFGS uses Frobenius norm: $\|A\|_{W_k}^2 = \|W_k^{1/2} A W_k^{1/2}\|_F^2 = \text{tr}(A W_k A^T W_k)$ w/ $W_k s_k = y_k$ i.e., putting move weight on directions by (s_k, y_k)

⇒ Final formulation

BFGS chooses $H_{k+1} = \arg\min_{H \in \mathbb{H}^n, H y_k = s_k} \|H - H_k\|_{W_k}$ i.e., $\begin{cases} H_{k+1} \text{ stays close to } H_k \text{ under weighted norm} \\ H_{k+1} \text{ remains sym.} \\ H_{k+1} \text{ satisfies inverse recant eq: } H_{k+1} y_k = s_k \end{cases}$

$= (I - \rho_k s_k y_k^T) H_k (I - \rho_k y_k s_k^T) + \rho_k s_k s_k^T$ where $\rho_k = \frac{1}{y_k^T s_k}$

Algorithm BFGS

```
1: for  $k = 1..T$  (or until  $\|\nabla f(x_k)\| < \epsilon\| \|$ ) do
2:    $x_{k+1} = x_k - \eta_k H_k \nabla f(x_k)$ 
3:   where
4:      $s_k = x_{k+1} - x_k$ 
5:      $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$ 
6:      $\rho_k = 1/y_k^T s_k$ 
7:      $H_{k+1} = (I - \rho_k s_k y_k^T) H_k (I - \rho_k y_k s_k^T) + \rho_k s_k s_k^T$ 
8: end for
```

작성 문제 출제

1-BFGS) Even BFGS is expensive! ($O(n^2)$)

⇒ Don't store H_k at all, just keep the last m pairs (s_k, y_k)

⇒ $O(mn) < O(n^2)$, Hcl quasi-Newton method for large-scale prob.

1-may: 1-BFGS vs. Newton

	Newton	BFGS / L-BFGS
Needs $\nabla^2 f$?	yes (analytic / autodiff)	no — only gradients
Matrix inversion?	yes, $O(n^3)/\text{step}$	no — update $H_k = B_k^{-1}$ directly
Per-step cost	$O(n^3)$	$> O(n^2)$ ($O(mn)$ for L-BFGS)
Memory	$O(n^2)$	$> O(n^2)$ ($O(mn)$ for L-BFGS)
Convergence	quadratic (best)	super-linear

Key principle: BFGS's update of B_k is rank-2.

Sherman–Morrison–Woodbury turns a rank-2 update of B_k into a rank-2 update of $H_k = B_k^{-1}$ — never needs to invert.

5. Dual Methods

Recap) primal prob: $p^* = \min_{x \in \mathbb{R}^n} \ell(x)$ s.t. $\ell_i(x) \leq 0, i=1,2,\dots,m, A x = b$

→ $L(x, \lambda, v) = \ell(x) + \sum_i \lambda_i \ell_i(x) + v^T (A x - b), \lambda \geq 0$

→ dual obj $g(\lambda, v) = \inf_x L(x, \lambda, v) : \text{ccv}$

& dual prob: $d^* = \max_{\lambda \geq 0, v} g(\lambda, v)$

→ duality gap: $p^* - d^* \geq 0$

→ Strong duality: $p^* = d^*$, holds under Slater's cdtn: $\exists \hat{x}$ w/ $\ell_i(\hat{x}) < 0, \forall i, A \hat{x} = b$

5.1. Why Duality Methods? : Motiv. for working in the dual

Bgnd) Why NOT GD to the primal?

B/c of the **constraint**. Constraint makes it hard to update x efficiently while staying feasible.

→ Dual gives us decomposability

$$\text{ex. } f(x) = \sum_i b_i(x_i) \rightarrow L(x, v) = \sum_i [b_i(x_i) + v^T A_i x_i] - v^T b$$

$$\rightarrow g(v) = \sum_i \inf_{x_i} [b_i(x_i) + v^T A_i x_i] - v^T b$$

(solve in parallel)
(dual prob. split into N ind. subproblems)

⇒ Push for this lecture

$$: \min_{x \in \mathbb{R}^n} b(x) \text{ s.t. } Ax = b \rightarrow \begin{cases} L(x, v) = b(x) + v^T (Ax - b) \\ g(v) = \inf_x [b(x) + v^T Ax] - v^T b \end{cases}$$

: Dual prob. $\max g(v)$ is always cvx

5.2. Dual Gradient Methods : Gradient ascent on dual prob

Bgnd) If b is **strictly cvx**, $x^* = \arg\min L(x, v)$ is **unique** for each v

& g is **diff.** w/ $\nabla g(v) = Ax^*(v) - b$: Constraint Residual (제약조건) 만이 남음!

Dual Gradient Ascent)

$$v^{k+1} = v^k + \alpha \nabla g(v^k) = v^k + \alpha (Ax^*(v^k) - b)$$

$$\text{ALGO iteration} \begin{cases} 1. \text{ Primal step : } x^k = \arg\min [b(x) + (v^k)^T Ax] \\ 2. \text{ Dual step : } v^{k+1} = v^k + \alpha (Ax^k - b) \end{cases}$$

$$\text{ex) } \min_x \frac{1}{2} x^T P x - q^T x \text{ s.t. } Ax = b, P \succ 0$$

$$\rightarrow \nabla L = 0 \Leftrightarrow x^*(v) = P^{-1}(q - A^T v)$$

$$\rightarrow g(v) = \frac{1}{2} x^{*T} P x^* - q^T x^* + v^T (Ax^* - b) \Leftrightarrow g(v) = -\frac{1}{2} v^T A P^{-1} A^T v + (A P^{-1} q - b)^T v - \frac{1}{2} q^T P^{-1} q$$

$$\rightarrow \nabla g(v) = Ax^*(v) - b = A P^{-1}(q - A^T v) - b, \nabla^2 g(v) = -A P^{-1} A^T \rightarrow \alpha = 1/\sigma \text{ w/ } \sigma = \|A P^{-1} A^T\|$$

$$\Rightarrow v^{k+1} = v^k + \alpha (A P^{-1}(q - A^T v) - b)$$

Thm) Let b be μ -strongly **cvx**, then g is **smooth** w/ $\sigma = \|A\|^2/\mu$.

Dual gradient ascent w/ step $\alpha = 1/\sigma$ satisfies

$$: d^* - g(v^k) \leq \sigma \cdot \frac{\|v^0 - v^*\|^2}{2k} = O(1/k)$$

BUT when b is **1-smooth**, effective ctn # = $\frac{1}{\mu} \cdot \frac{\|A\|^2}{\lambda_{\min}(AA^T)}$: very large if $\mu/1 < 1$ (ill-conditioned)

when b is **merely cvx** ($\mu=0$), g is **NOT smooth** $\Rightarrow$ subgradient 쓰지만 $O(1/\sqrt{k})$ 로 매우 느림

5.3. Augmented Lagrangian : Method of Multipliers

Bgnd) Limitations of dual gradient

1. b needs to be strongly **cvx** for g to be smooth
2. Cnvg rate $O(1/k)$ w/ $K = \|A\|^2/\mu$
3. Need exact sol. of x -subprob. at each step

$$\text{Def) } L_p(x, v) = b(x) + v^T (Ax - b) + \frac{\rho}{2} \|Ax - b\|^2 : \text{vanishes at feasible point}$$

의미: b 가 strongly **cvx** 아니어도 L_p 를 strongly **cvx** 하게 만들음. dual은 strongly **cvx** 하게 만들음

Algorithm Method of Multipliers

```

1: Choose  $\rho > 0$ , initialize  $\nu^0$ 
2: for  $k = 0, 1, 2, \dots$  do
3:    $x^{k+1} = \arg \min_x L_\rho(x, \nu^k)$ 
4:    $\nu^{k+1} = \nu^k + \rho(Ax^{k+1} - b)$ 
5: end for

```

Note: the dual step size equals ρ —it matches the smoothness of the augmented dual.

The name reflects step 3: the Lagrange multipliers ν are updated alongside the primal variable.

Intt)

$$L_\rho(x, \nu) = f(x) + \nu^T(Ax - b) + \frac{\rho}{2} \|Ax - b\|^2$$

$$= f(x) + \frac{\rho}{2} \|Ax - b + \frac{\nu}{\rho}\|^2 - \frac{\|\nu\|^2}{2\rho}$$

$$\Rightarrow x^{k+1} = \arg \min_x f(x) + \frac{\rho}{2} \|Ax - b + \frac{\nu^k}{\rho}\|^2$$

의 H = $\rho A^T A$ 2. ρ 커지면 (+ : stronger dual cavity, - : f 를 짓눌러 subprob (x 찾기) 가 어려워짐)

Thm)

Let f be cvx & have $b.c.v.$ then method of multipliers converges: $\|Ax^k - b\| \rightarrow 0, f(x^k) \rightarrow p^*$

When f is also strongly cvx , convergence is linear (geometric rate).

Dual Gradient 비한 장점: 1. f 가 strong cvx 아니어도 작동 2. Strong cvx $f \rightarrow$ linear cgc

Limitation)

$$x^{k+1} = \arg \min_x f(x) + \frac{\rho}{2} \|Ax - b + \frac{\nu^k}{\rho}\|^2$$

$\Rightarrow$ Even if $f = \sum_i f_i(x_i)$ is separable, penalty $\|Ax - b\|^2$ re-couples variables and ∇ parallelism (=decomposability) NOT

5.4. Alternating Direction Method of Multipliers

Def)

WTO $\min_{x,z} f(x) + g(z)$ s.t. $Ax + Bz = c$

w/ $f, g: cvx, A \in \mathbb{R}^{n \times n}, B \in \mathbb{R}^{n \times m}, c \in \mathbb{R}^n$

$$\Rightarrow \text{Augmented Lagrangian: } L_\rho(x, z, \nu) = f(x) + g(z) + \nu^T(Ax + Bz - c) + \frac{\rho}{2} \|Ax + Bz - c\|^2$$

x, z 한꺼번에 Min. 하지 않고 ∇ 개별 개별로 접근

Algorithm ADMM

```

1: Initialize  $z^0, \nu^0$ 
2: for  $k = 0, 1, 2, \dots$  do
3:    $x^{k+1} = \arg \min_x f(x) + \frac{\rho}{2} \|Ax + Bz^k - c + \frac{\nu^k}{\rho}\|^2$ 
4:    $z^{k+1} = \arg \min_z g(z) + \frac{\rho}{2} \|Ax^{k+1} + Bz - c + \frac{\nu^k}{\rho}\|^2$ 
5:    $\nu^{k+1} = \nu^k + \rho(Ax^{k+1} + Bz^{k+1} - c)$ 
6: end for

```

시험나옴, 식 전체 작성

Steps 1 and 2 each involve only f or g respectively—they can be solved independently.

Rebline dual var. $u = \nu/\rho$

$$\text{then } u^{k+1} = u^k + \frac{Ax^{k+1} + Bz^{k+1} - c}{\rho}$$

$\rightarrow$ At convergence, $Ax^* + Bz^* = c$ & u^* : constraint price

Method	When to use	Convergence	Decomposable	Key requirement
Dual gradient	simple, strongly convex f	$O(1/k)$	✓	f strongly convex
Method of Multipliers	fast convergence needed	$O(1/k)$ / linear [†]	✗	full-rank A
ADMM	large-scale, distributed	$O(1/k)$ / linear [†]	✓	separable structure

[†]Linear convergence requires stricter conditions: f (or g for ADMM) must be both strongly convex and smooth, and the constraint matrices must have full column rank; otherwise $O(1/k)$.

- ▶ All three methods are gradient ascent on some dual—they differ in how the dual is constructed
- ▶ ADMM is the method of choice for large-scale structured problems
- ▶ Next lecture: distributed optimization in more depth

Prop)

a) **Decomposable**: x & z only include f & g

b) **Flexible**: f, g can be non-smooth, can include indicators (for hard constraint) or have old-form proximal updates

c) **Scalable**: each subprob < original prob

d) **Practical**: works well in practice

Thm)

Let f, g : cld cvx ftns & prob³ sol. Then ADMM converges

: Primal residual $r^k = Ax^k + Bz^k - c \rightarrow 0$ Rate: $O(1/k)$ under mild cdtns

Dual residual $s^k = \rho A^T(Bz^k - z^{k-1}) \rightarrow 0$

Obj. $f(x^k) + g(z^k) \rightarrow p^*$

Penalty ρ trades off cgc / speed

5.5. Applications : Lasso & Distributed Opt.

5.5.1. Lasso

Def) $\min_{x \in \mathbb{R}^n} \frac{1}{2} \|Ax - b\|^2 + \lambda \|x\|_1$ *NOT smooth*

→ ADMM form) $\min_{x, z} \frac{1}{2} \|Ax - b\|^2 + \lambda \|z\|_1, \text{ s.t. } x = z$

→ $x^{k+1} = \arg \min_x \frac{1}{2} \|Ax - b\|^2 + \frac{\rho}{2} \|x - v\|^2, \quad v = z^k - u^k$

$\nabla = 0 \rightarrow x^{k+1} = (A^T A + \rho I)^{-1} (A^T b + \rho v)$

→ $z^{k+1} = \arg \min_z \lambda \|z\|_1 + \frac{\rho}{2} \|z - v\|^2, \quad v = x^{k+1} + u^k$

∵ z 대한 목적함수가 j 로 쪼개기 가능 $\rightarrow \min_{z_j} \lambda |z_j| + \frac{\rho}{2} (z_j - v_j)^2$
pull z toward 0 pull z toward v

Algorithm Lasso via ADMM

1: Choose $\rho > 0, \lambda > 0$; initialize $z^0 = 0, u^0 = 0$
 2: **for** $k = 0, 1, 2, \dots$ **do**
 3: $x^{k+1} = (A^T A + \rho I)^{-1} (A^T b + \rho(z^k - u^k))$
 4: $z^{k+1} = \frac{S_{\lambda/\rho}(x^{k+1} + u^k)}{S_{\lambda/\rho}(x^{k+1} + u^k)}$
 5: $u^{k+1} = \frac{S_{\lambda/\rho}(x^{k+1} + u^k)}{S_{\lambda/\rho}(x^{k+1} + u^k)}$
 6: **end for**

The matrix $(A^T A + \rho I)$ is positive definite and can be factored once before the loop; each iteration then requires only a linear solve and a soft threshold.

5.5.2. Distributed Opt.

Def) $\min_{x \in \mathbb{R}^n} \sum_i b_i(x)$: global prob w/ N agents having local data & local obj. b_i

→ $\min_{x_1, \dots, x_N, z} \sum_i b_i(x_i)$ s.t. $x_i = z_i \Leftrightarrow$ ADMM form w/ $b(x_1, \dots, x_N) = \sum_i b_i(x_i)$: separable & $g(z) = 0$

⇒ Augmented Lagrangian : $L_p = \sum_i b_i(x_i) + \sum_i [\bar{u}_i^T (x_i - z) + \frac{\rho}{2} \|x_i - z\|^2]$

⇒ Updates $\left\{ \begin{array}{l} \text{local update} : x_i^{k+1} = \arg \min_{x_i} b_i(x_i) + \frac{\rho}{2} \|x_i - z^k + u_i^k\|^2 \\ \text{Global avg (coordinator)} : z^{k+1} = \frac{1}{N} \sum_i (x_i^{k+1} + u_i^k) \end{array} \right.$

$\left\{ \begin{array}{l} \text{Dual update (at each agent } i) : u_i^{k+1} = u_i^k + x_i^{k+1} - z^{k+1} \end{array} \right.$

7차/적으로

6. Distributed OPT

6.1 Parallel SGD : The naive baseline : Avg gradients across workers

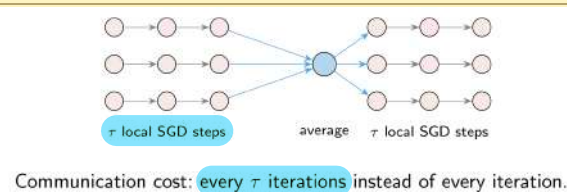
Bgnd) WTM minimize $f(x) = \frac{1}{N} \sum_i b_i(x)$ BUT No machine sees all the data

Def) Each worker i computes a stochastic ∇ of its local loss b_i ; avg them & take one SGD step
 $x^{k+1} = x^k - \frac{\eta}{n} \sum_i g_i^k$ (requires $b = \frac{1}{N} \sum_i b_i$) where $g_i^k = \nabla F(x^k; \xi_{i,k})$: unbiased estimate of $\nabla b_i(x^k)$

Intt) Every step ends w/ a global sync. Communication cost scales w/ # of steps.

6.2 Local SGD : Take τ local steps before averaging

Idea)



Def) Each worker i runs SGD locally for τ steps

$$: x_i^{s+1} = x_i^s - \eta \nabla F(x_i^s; \xi_i^s) \quad s = 0, \dots, \tau-1 \quad \begin{cases} \tau = 1 : \text{Parallel SGD} \\ \tau > 1 : \text{FedAvg} \end{cases}$$

Then avg

$$: x^{k+1} = \frac{1}{n} \sum_{i=1}^n x_i^{(\tau)}$$

Prop) a) τ trade-offs : time / outer iter = $\tau \cdot t_{\text{comp}} + t_{\text{comm}}$
 $\begin{cases} \text{small } \tau : \text{accurate global step BUT } \nearrow \text{comm} \\ \text{large } \tau : \searrow \text{comm. BUT iterates drift apart} \end{cases}$
 $\hookrightarrow$ Rule of thumb : pick $\tau \propto \sqrt{t_{\text{comm}} / t_{\text{comp}}}$

b) When Local SGD (works : Comm. is expensive, Workers' data are i.i.d.
 (breaks down : Highly NON-i.i.d. data, many local steps amplify the disagreement)
 단점 설명

c) Fed. Learning = Local SGD w/ extras (ex. client sampling, secure aggregation, ...)
 $\hookrightarrow$ Core ALGO : FedAvg = Local SGD w/ $\tau \gg 1$

6.3. Decentralized SGD : When connection to a central server is unreliable

Def) Avg consensus : the building block

Each worker i holds x_i . Repeatedly mix w/ neighbours $N(i)$

$$: x_i^{t+1} = \sum_{j \in N(i) \cup \{i\}} W_{ij} x_j^t \quad \text{w/ mixing matrix } W \text{ (doubly stochastic)}$$

Under mild cdtns, $x_i^t \rightarrow \bar{x}$ as $t \rightarrow \infty$
 local avg global avg

Def) DSGD = gossip + gradient
 작성문제 출처

$$\Rightarrow x_i^{k+1} = \sum_{j \in N(i) \cup \{i\}} W_{ij} x_j^k - \eta \nabla F(x_i^k; \xi_i^k)$$

	Centralized	Decentralized
Coordinator	yes (server)	none
Comm pattern	all-to-one	neighbor-to-neighbor
Server load	$\mathcal{O}(n)$	—
Fault tolerance	weak (server)	strong
Convergence	fastest	topology-limited

Thm) Let $X^t \in \mathbb{R}^{n \times d}$ stack the workers' iterates as rows. Update $X^{t+1} = W X^t$ w/ doubly stochastic W satisfies
 $: \|X^t - \bar{1} \bar{x}^T\|_F \leq \rho(W)^t \|X^0 - \bar{1} \bar{x}^T\|_F$

where $\rho(W) = |\lambda_2(W)|$ is the second-largest eigenvalue of W in abs. value (the spectral gap)

hw2 #5

Prob) W is sym. doubly stochastic $\Rightarrow$ eigenvalues

$$1 = \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq -1 ; \text{top eigenvector } v_1 = \bar{1} / \sqrt{n}$$

1. Decompose $X^0 = \bar{1} \bar{x}^T + \varphi^0$ where $\varphi^0 \perp \bar{1}$

2. Mixing preserves the mean : $W \bar{1} = \bar{1}$, $W X^0 = \bar{1} \bar{x}^T + W \varphi^0$

3. On the orthogonal complement of $\bar{1}$, W acts w/ spectral radius $\rho(W) = |\lambda_2(W)| : \|W \varphi^0\|_F \leq \rho(W) \|\varphi^0\|_F$

4. Apply t times : $\|X^t - \bar{1} \bar{x}^T\|_F \leq \rho(W)^t \|X^0 - \bar{1} \bar{x}^T\|_F$

Prop) (+) : NO bottleneck ($\mathcal{O}(1)$ /link), Fault tolerance ($\times$ single point of failure), Privacy

(-) : Cost (slow convergence)

7. Non-cvx OPT & Advanced Topics

Actually Wrap-up, Mostly Recent ex.)

/moy)	Week & Thesis	Key Concept	Used In
	GD	Convergence Analysis, step size	All papers
	Adaptive	Adam, coordinate-wise scaling	Basis Rotation
	Second-Order	Hessian, preconditioning	SASSHA
	Dual / ADMM	Lagrangian, Duality, ADMM	GEM, ELSA
	Distributed	Async Updates	Basis Rotation

중 3개씩 이름/설명.

기말 복기

1. 말문제 생각보다 많음 4음 $\Rightarrow$ 각 방법 분류 (ex ADMM, Dual) & 전체 흐름 파악이 중요.
2. ω 2 #1이 #1로 나옴 (살짝 변형) $\Rightarrow$ 주어진 자료는 풀이법 숙지해두기
3. 뒷부분 (Quasi-Newton ~)이 비중이 큼. GD, Accelerated Method 거의 안 나오니 좀 더 준비해도 될 듯.